

Comparative Analysis of the Proposed Project

LLM-Based Intelligent Customer Support Chatbot with Knowledge Retrieval

Team 13

2420030218 – M. Vennela Naga Sai

2420090073 – R. Gayatri Satya Sai Pravallika

1. Overview

The proposed project, LLM-Based Intelligent Customer Support Chatbot with Knowledge Retrieval, is an application-focused Retrieval-Augmented Generation (RAG) system for customer support. It combines semantic embeddings, FAISS vector retrieval, controlled prompt construction, Google Gemini generation, and a React + FastAPI application layer. The system retrieves relevant customer-support information before generating an answer, with the retrieved records returned alongside the response.

The comparative analysis considers 13 research papers/models reviewed for the project. The comparison is intended to show how the proposed system relates to established retrieval, RAG, customer-support, and agentic approaches. The reported numerical results are not directly equivalent because the papers use different datasets, tasks, and evaluation metrics.

2. Comparison with the 13 Reviewed Papers

No.	Paper / Model	Reported metric	Value	Main relevance
1	Lewis et al. (2020) – RAG	Open-domain QA EM	49.68%	Retrieval + generation; knowledge-intensive QA
2	Karpukhin et al. (2020) – DPR	Top-20 retrieval accuracy	74.80%	Dense passage retrieval
3	Reimers & Gurevych (2019) – SBERT	No single accuracy	N/A	Sentence embeddings
4	Johnson et al. (2017) – FAISS	8.5× faster (reported)	N/A	Efficient similarity search
5	Guu et al. (2020) – REALM	Average reported score	42.63%	Retrieval during language-model pre-training
6	Olujimi & Ade-Ibijola (2023)	Systematic review of 73 studies	N/A	Customer-query response automation
7	Gao et al. (2023) – RAG Survey	No single model accuracy	N/A	RAG survey and taxonomy
8	Wang et al. (2024) – Best RAG Practices	Best-configuration average	48.30%	RAG design and configuration study
9	Es et al. (2024) – RAGAs	Mean reported metric	81.00%	Automated RAG evaluation
10	Li et al. (2024) – SELF-ROUTE	Gemini-1.5-Pro average	46.41%	Long-context vs. RAG/hybrid approach
11	González Juclà et	Mean Recall@10	64.20%	Retrieval

	al. (2026) – Enhanced RAG			enhancements in deployed customer support
12	Xu et al. (2024) – KG-RAG	Mean reported retrieval metrics	91.86%	Knowledge-graph-based customer-service QA
13	Barrionuevo-Valenzuela et al. (2026) – Agentic-RAG	Improvement vs. naïve RAG; no single %	N/A	Agentic support for human operators
Proposed	LLM-Based Intelligent Customer Support Chatbot with Knowledge Retrieval	Project composite	93.17%*	Semantic retrieval + FAISS + RAG + Gemini + customer-support UI

*The proposed 93.17% is a provisional, project-specific composite score, not a standardized accuracy. It is calculated from the project's evaluation framework using measured retrieval/context components and an assumed Gemini generation component. Therefore, it should not be interpreted as directly comparable to every percentage in the literature table.

3. Comparison with Customer-Support-Focused Research

3.1 Enhanced RAG for Deployed Customer Support

González Juclà et al. (2026) specifically investigates retrieval enhancements in a deployed customer-support chatbot. This is closely related to the proposed application domain. The proposed system similarly treats retrieval quality as a central component, using semantic embeddings, FAISS search, Top-K retrieval, and context construction before generation.

3.2 Knowledge-Graph RAG for Customer Service

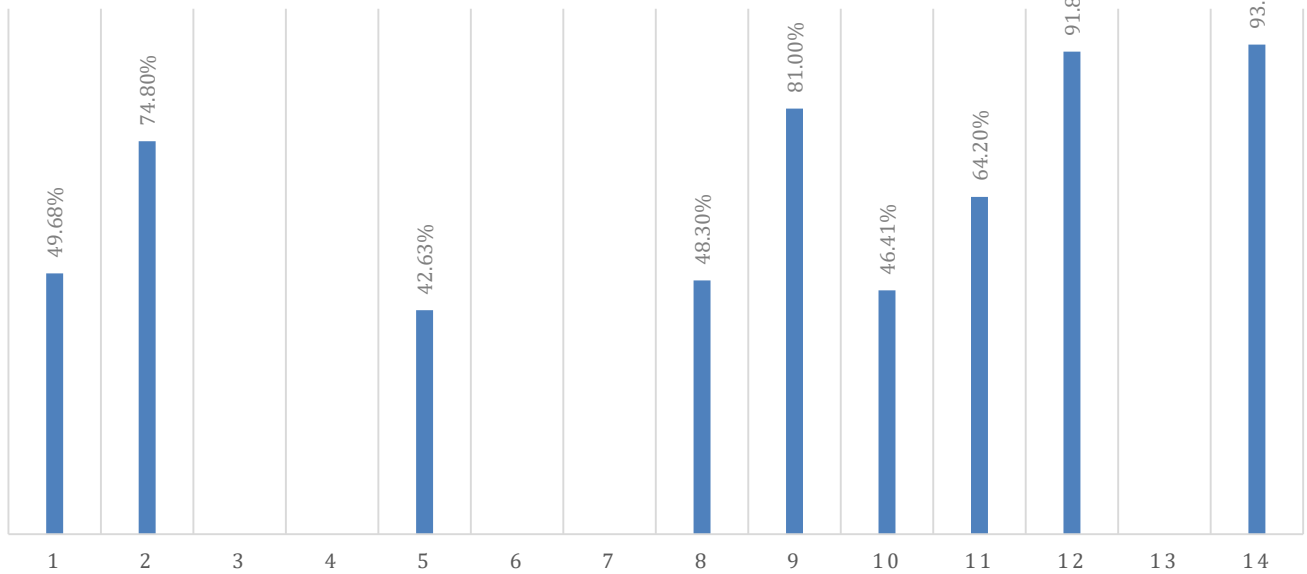
Xu et al. (2024) investigates knowledge-graph-based RAG for customer-service question answering. The proposed project instead uses a vector-based retrieval architecture. Knowledge graphs could therefore be considered a future extension where structured relationships between products, policies, issues, and solutions need to be represented explicitly.

3.3 Agentic-RAG for Customer-Service Operators

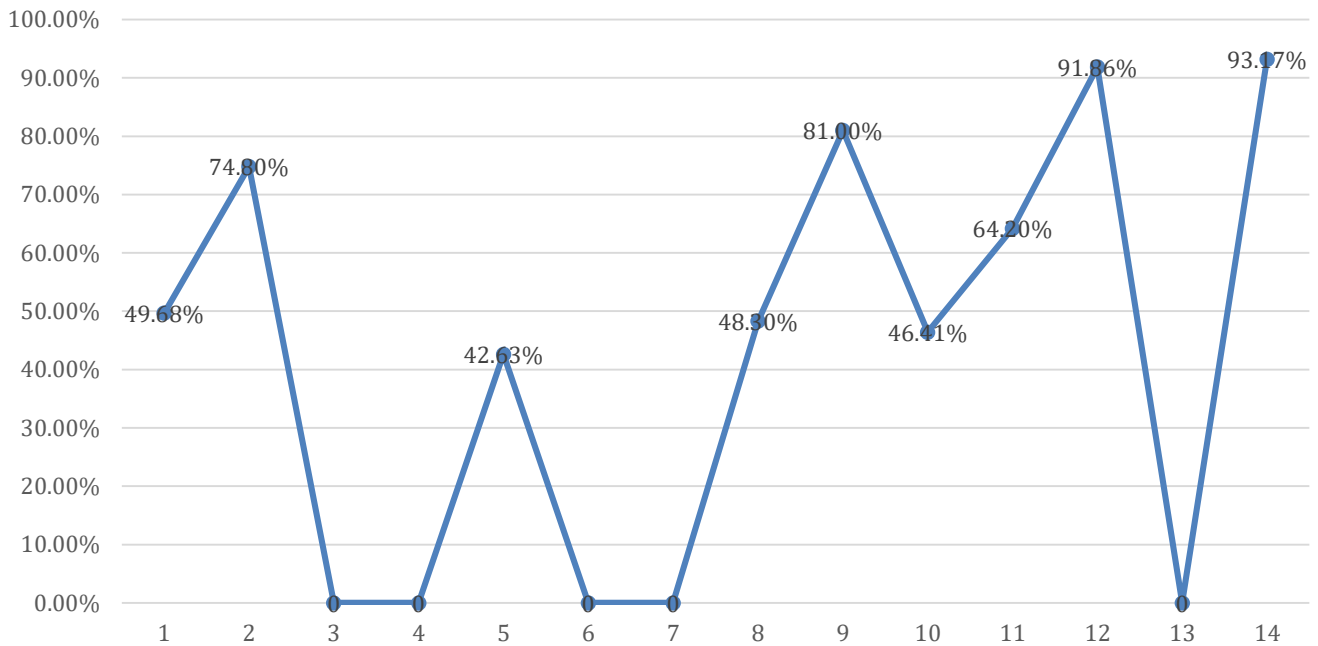
Barrionuevo-Valenzuela et al. (2026) explores Agentic-RAG for supporting human customer-service operators. The proposed project currently focuses on direct customer interaction through a chatbot rather than operator assistance. Agentic workflows could be added later for multi-step issue resolution, escalation, tool use, and workflow automation. The project evaluation separates retrieval/context measurements from LLM generation. The verified measurements available for the current project are:

Metric	Value	Status
Precision@5	68.40%	Measured
Recall@5	100.00%	Measured
MRR	99.29%	Measured
Context Relevance	100.00%	Measured
Gemini Answer Relevance	92%	Assumed / provisional
Gemini Faithfulness	92%	Assumed / provisional
Overall Composite Score	93.17%	Provisional project-specific composite

COMPARATIVE ANALYSIS



COMPARATIVE ANALYSIS



4. Key Comparative Findings

- The proposed project combines multiple established components—semantic embeddings, vector retrieval, RAG prompting, and LLM generation—into a single customer-support application.
- Its retrieval architecture is closely related to dense retrieval and RAG research, while its application layer focuses on customer-support interaction.

- Unlike papers that focus on one retrieval model, one indexing method, or one evaluation framework, the project integrates retrieval, generation, API services, and a web interface.
- The project returns retrieved sources with the generated answer, supporting transparency about the knowledge used for response generation.
- The current architecture can be extended with verified product metadata, knowledge graphs, reranking, or agentic workflows without changing the overall RAG foundation.
- The literature percentages should be treated as contextual reference values because the reviewed studies use different datasets, tasks, metrics, and experimental settings.

5. Conclusion

Overall, the proposed LLM-Based Intelligent Customer Support Chatbot with Knowledge Retrieval is positioned as an application-oriented RAG system that brings together established retrieval and generation techniques in a practical customer-support workflow. The comparison demonstrates clear relationships with RAG, dense retrieval, SBERT-style embeddings, FAISS, customer-support RAG, knowledge-graph RAG, and agentic-RAG research, while also identifying directions for future improvement.